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Installing a Modular Sensor in VMware

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M04 Modular Sensor Installation and Configuration on ESXi (01h:52m)

Introduction (06m:00s)

Get introduced to installing and configuring a Stellar Cyber Modular Sensor on VMware ESXi. Learn about log forwarding, network traffic features, and the role of the sensor in collecting and analyzing telemetry for enhanced security operations.

Overview (33m:00s)

Explore the high-level placement and capabilities of Modular Sensors within the Stellar Cyber Platform, including key features such as network traffic capture, sandboxing, and IDS. Review firewall communication requirements for seamless deployment, then watch a demo of configuring Modular Sensor profiles to manage network flow, log forwarding, and security data collection to optimize Interflow ingestion into the data lake.

ESXi Configuration (25m:00s)

Configure the ESXi environment to support Modular Sensors within the Stellar Cyber Platform, covering the creation of port groups, network switches, and virtual network adapters. Learn how to enable promiscuous mode and set up network interfaces for both management and traffic monitoring, then follow a demonstration that walks through each configuration step to facilitate traffic capture and sensor deployment.

Modular Sensor CLI Configuration (17m:00s)

Learn CLI commands to configure network settings, including IP addresses, gateway, hostname, and DNS for Modular Sensors. Follow along with a hands-on demo showing each configuration step, then discover how to bind the sensor to the Stellar Cyber Platform Data Processor to establish connectivity and enable seamless data forwarding.

Modular Sensor Data Processor Configuration (12m:00s)

Discover how to authorize and configure the Modular Sensor within the Stellar Cyber Platform, ensuring secure communication between sensor and platform components. Watch a demonstration of the full authorization process, including how to activate profiles and enable log forwarding through the Stellar Cyber UI. Learn to monitor status indicators, confirm sensor health, and validate successful integration with the data lake to support threat intelligence and machine learning workflows.

Modular Sensor Configuration Validation (18m:00s)

Validate the Modular Sensor setup by monitoring syslog traffic and confirming that log data reaches the Stellar Cyber Platform for processing and analysis. Follow a demonstration to verify data flow between the Modular Sensor and the Stellar Cyber Platform, and learn techniques for checking log counters to ensure that sensor output is correctly forwarded and ingested into the data lake.

The first time you access a link on the portal during a session, you must log in to access content.

This topic describes how to install a Modular Sensor in a VMware environment using the installation file downloaded from the **Download Images** tab in the **System | DATA SOURCE MANAGEMENT | Sensors | Sensor Installation** page. Refer to the following sections for details:

- **About Modular Sensors**
- **Site Preparation**
- **Obtaining the Installation File**
- **Creating a Virtual Machine Port Group**
- **Installing the Modular Sensor Image**
- **Applying a Token to the Installed Sensor**
- **Configuring a Static IP Address (Optional)**
- **Configure NTP and Set the Timezone**
- **Enabling SSSE3 for the Modular Sensor VM**

Use our example as a guideline, as you might be using a different software version.

Stellar Cyber does not support the installation of third-party software on its virtual or physical device sensors.

About Modular Sensors

Sensors provide the data gathering foundation for Stellar Cyber's OpenXDR platform, gathering the right data with context. Modular sensors are purpose-built Stellar Cyber sensors that include both the host and the Stellar Cyber monitoring software. They are provided as both physical devices (Photon sensors) and virtual machine images for different target environments.

Previous releases provided a variety of different types of device sensors, including Network, Security, and Modular. Going forward, the only type of device sensor is Modular. You can use the Modular Sensor Profile to enable whatever sensor features you like, creating the same functionality provided by the different sensor types in previous releases.

A modular sensor lets you easily add the features you like to your sensor. This helps simplify your deployment and lets you manage the VM requirements for the sensors based on the modular features they use.

Modular Sensors always include log ingestion. From there, you can enable different features as part of your modular sensor profile:

- Enable the **Network Traffic** feature to monitor the virtual environment, the physical environment if connected to the span port of a physical switch, or the LAN segment via a mirror port on a switch. The sensor monitors network and server response times and can identify applications.

  The sensor converts that information to metadata and forwards it to the DP as Interflow. The DP can then provide security, DDoS, and breach attempt detections.
- Enable the **Sandbox** and **IDS** features to improve your security posture:
  - Sandbox lets you detect malware in files and network traffic through Stellar Cyber's integrated cloud service and also provides anti-virus services.
  - IDS lets you detect intrusion attempts using both files and network traffic.

Keep in mind that VM resource requirements increase as you add more features to the Modular Sensor Profile. Refer to **Virtual Modular Sensor Provisioning and Performance** for details on the resources required to run different combinations of features in a Modular Sensor Profile, as well as how to use the **show module** and **show module request** CLI commands to compare provisioned resources against those required to run specific feature combinations. Stellar Cyber only enables a Modular Sensor Profile on a sensor if the host VM's resources can support it.

Site Preparation

Refer to **Virtual Modular Sensor Provisioning and Performance** for details on the resources required to run different combinations of features in a Modular Sensor Profile. Provision your modular sensor according to the features that you plan on enabling.

You will also need:

- A token obtained from the DP that will manage the sensor.
- Open firewall ports for log ingestion.
- Open firewall ports for the features you plan on enabling in the Modular Sensor Profile for this sensor.

One physical port on the system will be configured to receive network traffic to be analyzed. It must have promiscuous mode capability and be connected to an appropriate data source.

In addition, the system BIOS must have VT-d/IOMMU enabled.

Supported ESXi Versions

Sensor installation is supported on the following ESXi versions:

- 8.0
- 7.0
- 6.7

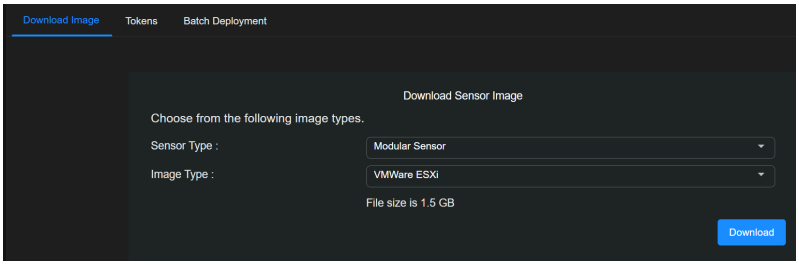
Obtaining the Installation File

You download the installation file for the Modular Sensor for VMware from the **Download Images** tab in the **System | DATA SOURCE MANAGEMENT | Sensors | Sensor Installation page**. Use the following procedure:

Only users with the **System | DATA SOURCE MANAGEMENT | Sensors | Sensor Installation | Sensor Image Download** privilege assigned to their profile in System | DATA SOURCE MANAGEMENT | Sensors | Sensor Installationthe System | ORGANIZATION MANAGEMENT | Role-Based Access Privileges interface can download images.

1. Navigate to the **System | DATA SOURCE MANAGEMENT | Sensors | Sensor Installation** page.
2. Set the **Sensor Type** dropdown to **Modular Sensor**.
3. Set the **Image Type** to **VMware ESXi**.

   The display updates to show you the size of the files to be downloaded.

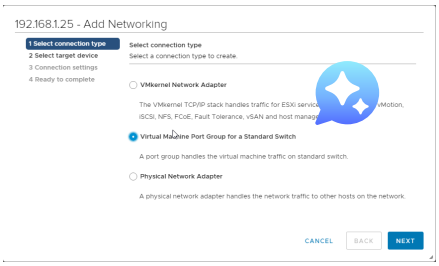


4. Click the **Download** button. The system downloads the installation file (aella-modular-ds-6.6.0.ova) along with its corresponding SHA-1 hash file.

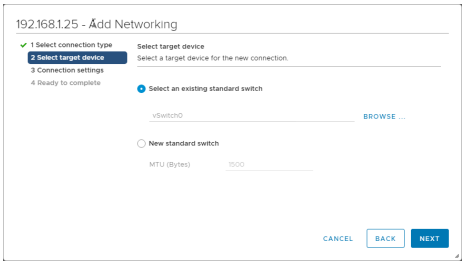
Creating a Virtual Machine Port Group

The following procedure creates a new virtual machine port group with port mirroring capabilities. This is the port group to which we'll connect the new modular sensor's monitor interface for traffic acquisition in the next procedure.

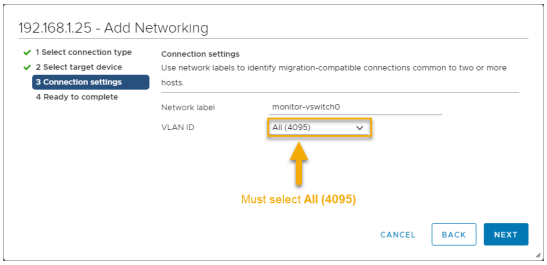
1. Log in to the vSphere Client, right-click the datacenter or host where you want to install the port group, and select the **Add Networking** option from the context menu that appears.
2. Select the Virtual Machine Port Group for a Standard Switch option and click Next.



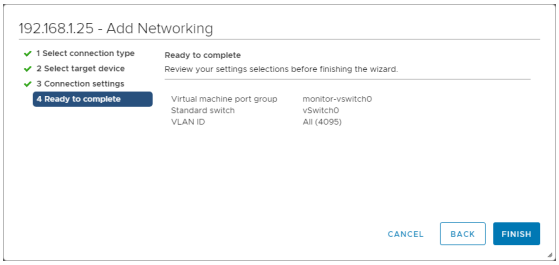
3. You can add the port group to an existing switch or create a new switch. The following image shows the attachment being made to an existing switch named **vSwitch0**.



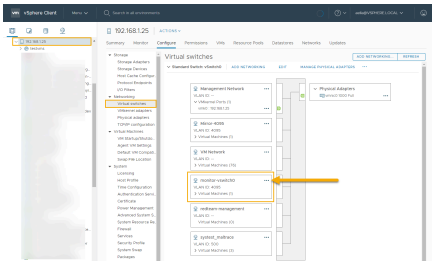
4. Create a network label with the VLAN ID of 4095, as shown in the image below:



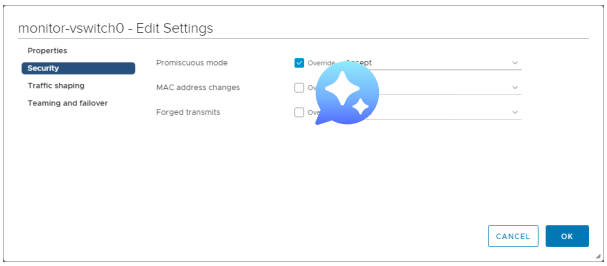
5. At this point the settings should be complete. Click the **Finish** button, as illustrated below.



6. Once setup is complete, you can view the new port group by right-clicking the host where it is installed, selecting **Settings** from the context menu that appears, and then selecting the **Networking | Virtual Switches entry**, as shown in the image below.



7. Click the three-dot icon for the port group and select **Edit Settings** from the context menu that appears.
8. Click on the **Security** entry in the left navigation panel and enable **Promiscuous Mode** by checking the corresponding **Override** and **Accept** boxes, as shown in the image below. Then, click **OK** to apply your settings.

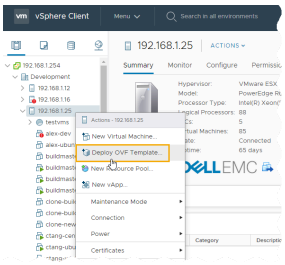


Installing the Modular Sensor Image

This section describes how to install the Modular Sensor image you downloaded from the DP at the start of the procedure.

Select the option to deploy a new OVF template wizard and use the **Local** file option, as shown below.

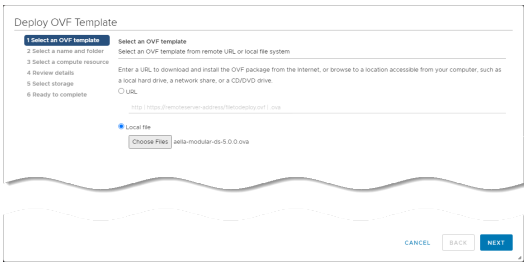
1. Log in to the vSphere Client.
2. Right-click the datacenter or host where you want to install the Modular Sensor and select the **Deploy OVF Template** option from the context menu that appears.



The Deploy OVF Template wizard starts.

Stellar Cyber provides the Modular Sensor image as an OVA file. The OVA file includes the necessary OVF file as a component.

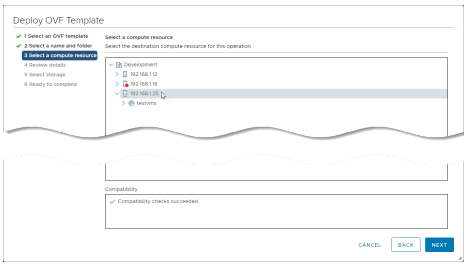
3. In the **Select an OVF Template** step, choose the **Local file** option and use the **Choose Files** button to navigate to the OVA file you downloaded in **Obtaining the Installation File**. Once you have selected the OVA file, click **Next**.



4. In the **Select a name and folder** step, provide a name for the virtual machine and select the data center where it will be deployed, as shown below. Then, click **Next** to continue.



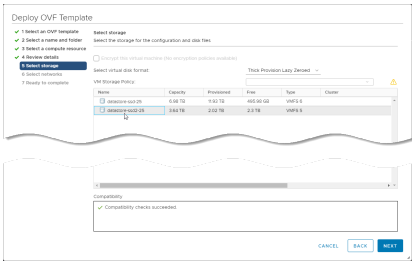
5. In the **Select a compute resource** step, select the resource that will run the virtual machine. This should be same resource where the mirror port is hosted. Then, click **Next** to continue.



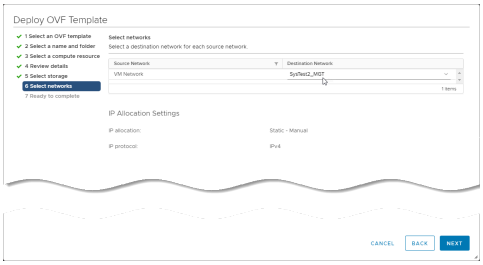
6. Review the details for the template and click **Next** to continue.



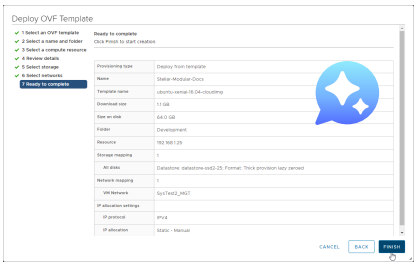
7. In the **Select storage** step, select the datastore to be used for the VM's configuration and disk files and click **Next** to continue.



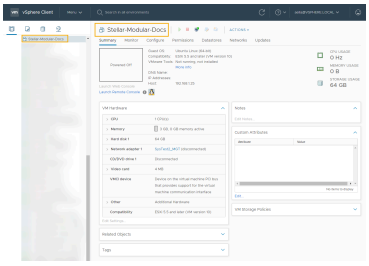
8. In the **Select networks** step, use the **Destination Network** dropdown to select the network where the Modular Sensor's management interface connects. Select a network that can access the DP that will manage the sensor and click **NEXT** to continue.



9. Review the settings in the **Ready to complete** step and click **FINISH** to start the deployment.



10. The deployment begins. You can keep tabs on its progress in the **Recent Tasks** panel at the base of the vSphere Client display. When the deployment completes, you can select its entry in the left navigation panel to see its settings. For example, in the image below, we have selected the Modular Sensor named **Stellar Modular Docs**.

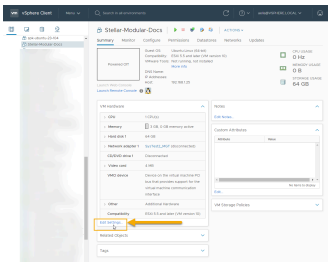


Adding a Monitoring Interface and Connecting it to the Port Group

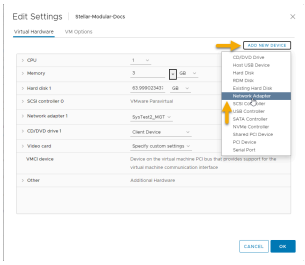
The next step is to add a second network adapter to the Modular Sensor virtual machine to be used as a monitor interface. This adapter will monitor traffic on the port group we created in [Creating a Virtual Machine Port Group](#).

You can only add a network adapter to the VM while it is powered off.

1. You should still be in the vSphere Client with the Modular Sensor virtual machine selected. Click the **Edit Settings** link in the **VM Hardware** panel.

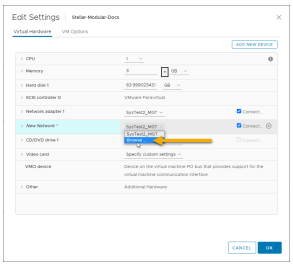


2. Click the **Add New Device** button and select **Network Adapter** from the context menu that appears.



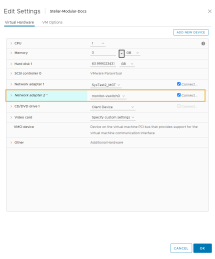
A new entry appears in the Edit Settings dialog box for **New Network**.

3. Click the dropdown for the new adapter's network and select the **Browse** option.



4. Select the entry for the port group you created in **Creating a Virtual Machine Port Group** and click OK.

   The new network adapter appears in the Edit Settings dialog box connected to the selected port group.



5. Make sure the **Connect** box for the new network adapter is checked, as illustrated above, and then click **OK** on the Edit Settings dialog box.
6. At this point the Modular Sensor is installed. If you need to provision any additional resources to support the modular features you anticipate enabling in the sensor's Sensor Profile, now is a good time to do so while it is still powered off. Then, you can start the VM.

Applying a Token to the Installed Sensor

The next step is to obtain and apply the token used to authorize and configure the installed sensor.

Obtaining a Token for the Installation

Tokens are required to authorize and configure the installation of a sensor image downloaded from the DP in the **System | DATA SOURCE MANAGEMENT | Sensors | Sensor Installation page**. Tokens point the installed sensor to the correct DP, assign the specified tenant, optionally provision a selected sensor profile, and authorize the sensor installation.

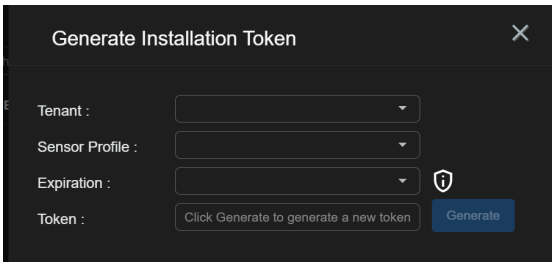
Use the following procedure to obtain a token in the Tokens tab:

1. Navigate to the **System | DATA SOURCE MANAGEMENT | Sensors | Sensor Installation** page and click on the **Tokens** tab.
2. If there is already an unexpired token that you want to use for the installation, you can either use the **Copy** button to copy it to the clipboard or use the **Download** button to download it as a file.
   - Copy the token if you plan on applying it by pasting it into a `set token string <token>` command in the CLI.
   - Download the token as a file if you plan on hosting the file on an HTTP server and referring to it in a `set token url <token url>` command.

Refer to **Assigning Tokens** for a summary of the different ways in which tokens can be applied to a sensor installation.

3. You can also click the **Generate** button to create a new token.

   The Generate Installation Token dialog appears:



4. Select the tenant for the token from the **Tenant** dropdown. This is the tenant to which all sensors authorized with this token will be automatically assigned. The dropdown lists all tenants configured for your organization in the **System | ORGANIZATION MANAGEMENT | Tenants** page.

5. Select the Sensor Profile to be assigned to all sensors authorized with this token from the **Sensor Profile** dropdown. The dropdown lists all sensor profiles available in the System | DATA SOURCE MANAGEMENT | Sensors | Sensor Profiles page for the selected Tenant.

   If you do not want to assign a Sensor Profile with a token, you can set this field to **None (no sensor profile selected)**. This is also the setting for any tokens migrated from a pre-5.3.0 release as part of an upgrade.

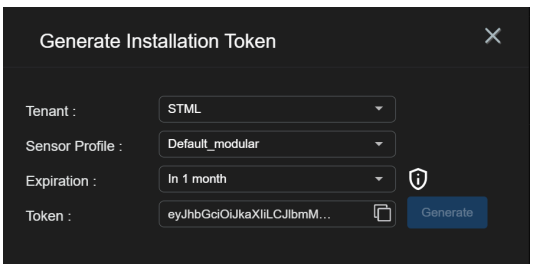
6. Use the **Expiration** dropdown to select an expiration date for this token. You can select specified expiration dates ranging from two weeks to three months.

   You can also select **Never expires** for the expiration date. However, Stellar Cyber recommends that you specify expiration dates for your tokens in order to improve the security of your deployment.

   Remember that when a token expires, **sensors authorized with the token continue to operate as normal**. Once a sensor successfully registers with the Stellar Cyber platform, it no longer uses the token. It is only used for the initial authorization, registration, and configuration of the sensor.

7. Click the **Generate** button.

   The system generates the token and displays its contents in the **Token** field. The dialog also updates to display the expiration date for the token, as illustrated below.



8. You can use the **Copy** button to copy the token to the clipboard immediately, or simply close the dialog and retrieve the token from the **Tokens** tab later on.

Applying the Token to the Sensor

Tokens are required to complete the installation of a sensor image downloaded from the DP in the **Download Image** tab.

You apply tokens to sensors as the last step in the overall installation procedure:

1. Log in to your new Sensor. The default username/password is **aella**/**changeme**. You are immediately prompted to change the password.
2. Apply the token to the installed sensor from the sensor CLI with the `set token` command using **one** of the options in the table below:



You only need to use **one** of the options in the table below. These are just two different ways to do the same thing – apply the token.

Option 1. Copy and Paste the Token String	Copy the token string from the **Tokens** tab and paste it into the CLI command. The syntax is as follows: `set token string <pasted string>`

Option 2. Host the Token on an HTTP Server	Download the token as a file from the **Tokens** tab, upload it to an HTTP server, and reference it in the `set token` command. The syntax is as follows:
	`set token url http://<url to token>`
	You can also use an HTTPS server. In that case, the specified URL must also include the username and password for the server using the following syntax:
	`set token url https://<user:password>@URL>`

3. The CLI reports that the **Sensor token is successfully set**.



If you receive an error message instead, it's possible that the token has expired. Refer to the **Tokens** tab to see the expiration date. If you are using the File technique, it's also possible that an extra space or line may have crept into your text file – check the file to make sure it includes only the token text.

4. Wait a minute or so. Then, verify that the token was successfully applied using any combination of the following techniques:
    - Check the **System | DATA SOURCE MANAGEMENT | Sensors | Sensors** list in the user interface to see that the sensor has registered itself successfully.
    - Verify that the `show system` command shows all services as running.
    - Verify that the `show receiver` command displays a receiver.
    - Verify that the `show json` command reports some data sent in the `BYTE_SENT` column.

Configuring a Static IP Address (Optional)

By default, the sensor uses DHCP for the management port's IP address. For ease of troubleshooting, however, Stellar Cyber recommends that you reconfigure the management port to use a static IP address. The procedure is as follows:

1. Log in to your sensor. The default username/password is **aella**/**changeme**, but you changed this when you applied the token in the previous section.
2. You can set IP parameters manually using commands similar to the following (substitute your own IP parameters for the ones shown in bold below):

   `set interface management ip` **`192.168.14.100/255.255.255.0`**

   `set interface management gateway` **`192.168.14.1`**

   `set interface management dns` **`8.8.8.8`**
3. Verify the IP settings with the `show interfaces` command.
4. Log out with the `quit` command.

Configure NTP and Set the Timezone

Stellar Cyber strongly recommends that configure NTP and set the timezone for the sensor.

Refer to **Best Practices for NTP and Timezones** for details.

Enabling SSSE3 for the Modular Sensor VM

The Modular Sensor VM must have SSSE3 enabled for its processors in order to operate correctly. In most cases, SSSE3 will already be enabled. However, if you encounter issues with packet collection or Interflow data generation, you can use the instructions below to ensure that SSSE3 is enabled.

Click here to see instruction on how to enable SSSE3.

Sensors installed on Linux hosts must have SSSE3 enabled for their processors in order to operate correctly. This is true for Modular Sensors and Linux Server Sensors, as well as legacy Network and Security Sensors.
SSSE3 is typically supported/enabled for most vCPUs, but may not be for certain legacy AMD vCPUs. See below for instructions on enabling SSSE3.

To enable SSSE3 for a virtual machine:

1. Start the virtual shell (virsh)
2. Type the following command to edit the virtual machine's settings:

   `edit <virtual_machine_name>`
3. Locate the **<cpu>** section. It should appear similar to the following:

```
<cpu mode='custom' match='exact' check='partial'>
    <model fallback='allow'>SandyBridge</model>
......
 </cpu>
```

4. Add the following line to the **<cpu>** section:

   ```
   <feature policy='require' name='ssse3'/>
   ```

   When you are done, the **<cpu>** section should appear similar to the following:

   ```
   <cpu mode='custom' match='exact' check='partial'>
       <model fallback='allow'>SandyBridge</model>
       <feature policy='require' name='ssse3'/>
   ...........
     </cpu>
   ```

5. Save changes to the virtual machine and exist virsh.
6. Run the following command:

   ```
   virsh define /etc/libvirt/qemu/<virtual_machine_name>.xml
   ```

   Each virtual machine has a configuration .xml file. Typically, these files are stored under **/etc/libvirt/qemu**, but the location may be different for your system.

7. Stop and start the virtual machine in virsh.

Preventing Clock Drift for Stellar Cyber VMs Deployed in ESXi

Virtual machines running on VMware ESXi can experience significant clock drift if the hypervisor's clock is not properly configured or if NTP synchronization on the ESXi host is unavailable. To help prevent clock drift on Stellar Cyber virtual machines deployed in ESXi, follow the guidance in Disabling Time Synchronization for Virtual Machines on the Broadcom Support site to disable one-off time synchronization for your Stellar Cyber VMs by adding the following advanced configuration parameters:

```
time.synchronize.continue = "FALSE"
time.synchronize.restore = "FALSE"
time.synchronize.resume.disk = "FALSE"
time.synchronize.shrink = "FALSE"
time.synchronize.tools.startup = "FALSE"
time.synchronize.tools.enable = "FALSE"
time.synchronize.resume.host = "FALSE"
```


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